



AESOP XI: Autonomous Edge-Computing Architecture Blueprint

Heterogeneous System Orchestration, Asymmetric Routing, & Autonomous Agent Flywheels

Specification Version: v2.2-Production-Ready | Target: Multi-Node
Sovereign Mesh

1. Canonical Build Sequence & Execution Roadmap

Phase 1: Grill with Docs Master Architectural Planning & Repo Partitioning

Phase 2: Global Knowledge Curation, File Parsing, & Universal JSONL Marker Indexing

Phase 3: Tool Harness Ingestion (Pocock, Honey for Devs, Prime Agent, ECC, GSD)

Phase 4: 3-APK Engine Deployment (Horizons UI + Shell Daemon + Media Daemon)

Phase 5: Local Dual-Model Inference & Continuous Improvement RLVR Loop (GCP)

Phase 6: Multi-Node Mesh Orchestration & Autonomous Agent Swarm (Jetson / Pi)

- **Priority 1: "Grill with Docs" Master Architectural Session & Strategic Roadmap**
 - **The Orchestration Driver:** Execute the master "Grill with Docs" session first to interrogate baseline documentation and establish the complete technical battle plan.
 - **Repository & Directory Partitioning:** Formulate the precise directory schemas, define strict repository isolation barriers, and assign individual agent manifests.
 - **Skill-to-Agent Allocation:** Map exactly which skills, harnesses, and tool endpoints are assigned to which agent runtimes (Node Alpha, Beta, Gamma, or Delta).
 - **Data Tiering Strategy:** Define explicit boundaries for what gets parsed, what stays on the active "hot plate" (mem0 context cache and local active reasoning ledgers), and what gets archived into the deep cold vault (Obsidian markdown vaults and Postgres OB1 vector stores).
- **Priority 2: Global Knowledge Curation & Universal JSONL Marker Indexing**
 - Assemble, clean, and partition core technical documentation: Qualcomm QAIRT/HTP SDK, Android Media/Accessibility APIs, Google Vertex AI Agent Builder, Llama Server, and Unsloth.
 - Attach dense JSONL marker schemes to every markdown file, tool repository, and skill guideline to allow direct KAG/RAG indexing without context bloat.
- **Priority 3: Tool Harness Ingestion & Action Schemas**
 - Transform static reference repositories into machine-executable skills.
 - Ingest developer harnesses: **Matt Pocock Skills** (Grill-Me, Spec, Ticket, TDD,

Code-Review), **Honey for Devs**, **Prime Agent** (RLM persistent REPL + Continual Harness with rollback snapshots), **ECC**, **GSD (Get Shit Done)**, **Claude Video**, **Reverse Skills**, **Code Review Graph**, **mem0**, and **OpenAI/Anthropic LocalAI Omni-Route bridges**.

- **Priority 4: 3-APK Native Android Deployment (Parallel with Priority 5)**
 - Deploy the decoupled 3-APK Android execution suite: **Horizons UI** (Orchestrator), **Shell Daemon APK** (Termux-free OS assistant/accessibility access), and **Media Daemon APK** (Screen-vision + Silero VAD + Speech stack).
- **Priority 5: Dual-Model On-Device Inference & Concierge Gateway**
 - Deploy the dual-model tandem: **Qwen 3.5 0.8B GGUF Q4_0** (Executor) and **Qwen 3.5 9B GGUF Q4_0** (Query / Metaprompt compiler).
 - Wire the on-device **Concierge Flow**: Voice capture $\rightarrow$ Silero VAD $\rightarrow$ Screen Vision context $\rightarrow$ Synthesized clean markdown meta-prompt $\rightarrow$ User verification $\rightarrow$ Frontier Model / LocalAI $\rightarrow$ Kokoro TTS speech output.
- **Priority 6: Network-Wide Integration & GCP Cloud Training Flywheel**
 - Establish the continuous improvement training flywheel: Real-time inference triples aggregated locally by Home Assistant $\rightarrow$ Packaged for batch audit $\rightarrow$ Validated by Red Agent $\rightarrow$ Exported to GCS $\rightarrow$ GCP Vertex AI / Compute Instances run RLVR (GRPO/Unsloth) fine-tuning $\rightarrow$ Updated weights pushed back to edge.
 - Activate the multi-agent home node: **Home Assistant Auditor & Housekeeper**, **Red Agent Batch Training Auditor**, **IT Help Desk & Manual Operator**, **Web Ingestion & News Monitor**, and **NPU Inference Manager**.

2. Multi-Node Compute Infrastructure

Node ID	Physical Platform	Primary Function	Core Acceleration & Compute	Primary Software & Daemons
NODE ALPHA	Moto Razr Ultra 2025	Local Orchestration & Concierge Hub	Snapdragon 8 Elite (Hexagon NPU v79)	3-APK Unified Stack (Horizons + Daemons), LocalAI, Moonshine/Kokoro
NODE BETA	Nvidia Jetson Orin Nano Super	Headless Home Server & Vector Hub	Dedicated CUDA Core Array (60–70 TOPs)	Postgres OB1 Protocol, Home Assistant, Red

				Agent Auditor, Multi-Agent Swarm
NODE GAMMA	Rubik Pi 3 (Dragonwing)	Display Workstation & Helpdesk Gateway	Thundercomm / Qualcomm SoC (14+ TOPs)	Dual-Monitor Display Engine, System Log Viewer, IT Manual Search
NODE DELTA	Google Cloud Platform	Asymmetric RAG & Heavy RLVR Training	Cloud TPUs / A100/H100 Instances	Vertex AI Agent Builder (\$1,000 Credit Tier), GCS Buckets, Unsloth/Axolotl GRPO

3. The 3-APK Unified Android Architecture





4. End-to-End System Dataflow & Runtime Routing





5. Multi-Agent Home Node Topology (Node Beta / Jetson)

- **Home Assistant (Cross-Agent Auditor & Housekeeper):** Manages real-time cross-agent auditing during live inference sessions, aggregates daily execution logs, compiles draft automation scripts, and packages raw telemetry pairs for downstream training.
- **Red Agent Auditor (Batch Training Guardrail):** Operates strictly when accumulated

context reaches the training compilation threshold, conducting air-gapped safety verification against the Nope Data Bank before datasets are shipped to GCP.

- **IT Help Desk & Manual Operator Agent:** Indexes local technical documentation, tool schemas, and hardware specs to debug operational failures and suggest system repairs.
- **Web Ingestion & News Monitor Agent:** Tracks breaking updates in local open-weight models, new tools, and upstream repository changes.
- **NPU / Local Inference Manager:** Balances weight offloading between Snapdragon Hexagon NPU, Jetson CUDA cores, and fallback cloud APIs.

6. GCP Cross-Account IAM Handshake & RLVR Loop

- **Business / Contractor Resource Tier:** Houses proprietary data in `gs://business-secure-vault-bucket/` and grants read-only roles/`storage.objectViewer` to the personal consumer service account.
- **Personal Developer Tier (\$1,000 Credit Pool):** Vertex AI Discovery Engine (`service-PERSONAL_PROJECT_NUMBER@gcp-sa-discoveryengine.iam.gserviceaccount.com`) points directly to the business bucket without copying data, absorbing 100% of RAG query costs.
- **The RLVR Training Loop:** Home Assistant packages verified task triples, Red Agent approves the batch, and the dataset streams to GCS. GCP instances run scheduled GRPO fine-tuning jobs on fixed base weights (Qwen/Llama) using Unsloth/Axolotl, returning fine-tuned GGUF weights back down to Node Alpha and Beta.

7. Decentralized Multi-Repository Map

```
└─ master_workspace/
├─ file-management-and-skills/ # Master Canonical Docs, JSONL Markers, Tool
  Schemas (Priority 1 Core)[cite: 1, 4]
├─ horizons-ui-v2.0/ # Kotlin/Java APK 1: Orchestration GUI, Webview, Chat
  Tile[cite: 8, 9]
├─ nova-daemon-shell/ # Kotlin/Java APK 2: OS Assistant & Accessibility
  Sandbox Bypass[cite: 8, 9]
├─ nova-daemon-media/ # Kotlin/Java APK 3: Screen Vision, Moonshine STT
  Kokoro TTS[cite: 1, 8, 9]
├─ agent-harness-hub/ # Prime Agent (RLM), Pocock Skills, Honey for Devs
  ECC, GSD[cite: 1, 7]
├─ localai-omni-route/ # Modality Routing, OpenAI/Anthropic Translation
  Endpoints[cite: 1, 6]
├─ obsidian-knowledge-vault/ # Graphify Networks, KAG/RAG Markdown Trees
  NotebookLM Hooks[cite: 1]
├─ red-agent-auditor/ # Air-gapped Batch Training Verification, Nope Data
  Bank[cite: 1]
```

```

└── node-beta-jetson/      # Postgres OB1 Vector Store, CUDA Inference
Servers[cite: 1]
└── gcp-training-flywheel/  # RLVR Export Scripts, Cross-Account IAM Handshake,
GRPO Pipelines[cite: 3, 5]

```

8. Universal JSONL Marker Schema

Every markdown documentation file and skill repository includes an attached .jsonl schema block to enable rapid, low-token search indexing across local agents and Vertex AI Discovery engines:

```

JSON
{
  "record_id": "SKILL_POCKOCK_GRILL_001"
  "document_path": "agent-harness-hub/pockock-skills/grill-me.md"
  "category": "SKILL_HARNESS"
  "target_runtime": "PRIME_AGENT_RLM"
  "metadata": {
    "title": "Grill-Me Architectural Interview Harness"
    "description": "Interactive question-driven prompt for technical planning"
    "primary_tools": ["claude-code", "open-wiki-cli", "prime-agent"]
    "required_context_keys": ["repo_scope", "target_architecture"]
  }
  "retrieval_tokens": ["grill", "spec", "architecture", "audit", "pockock"]
  "entry_points": {
    "repl_command": "/skill run grill-me"
    "jsonrpc_method": "agent.skills.execute"
  }
}

```